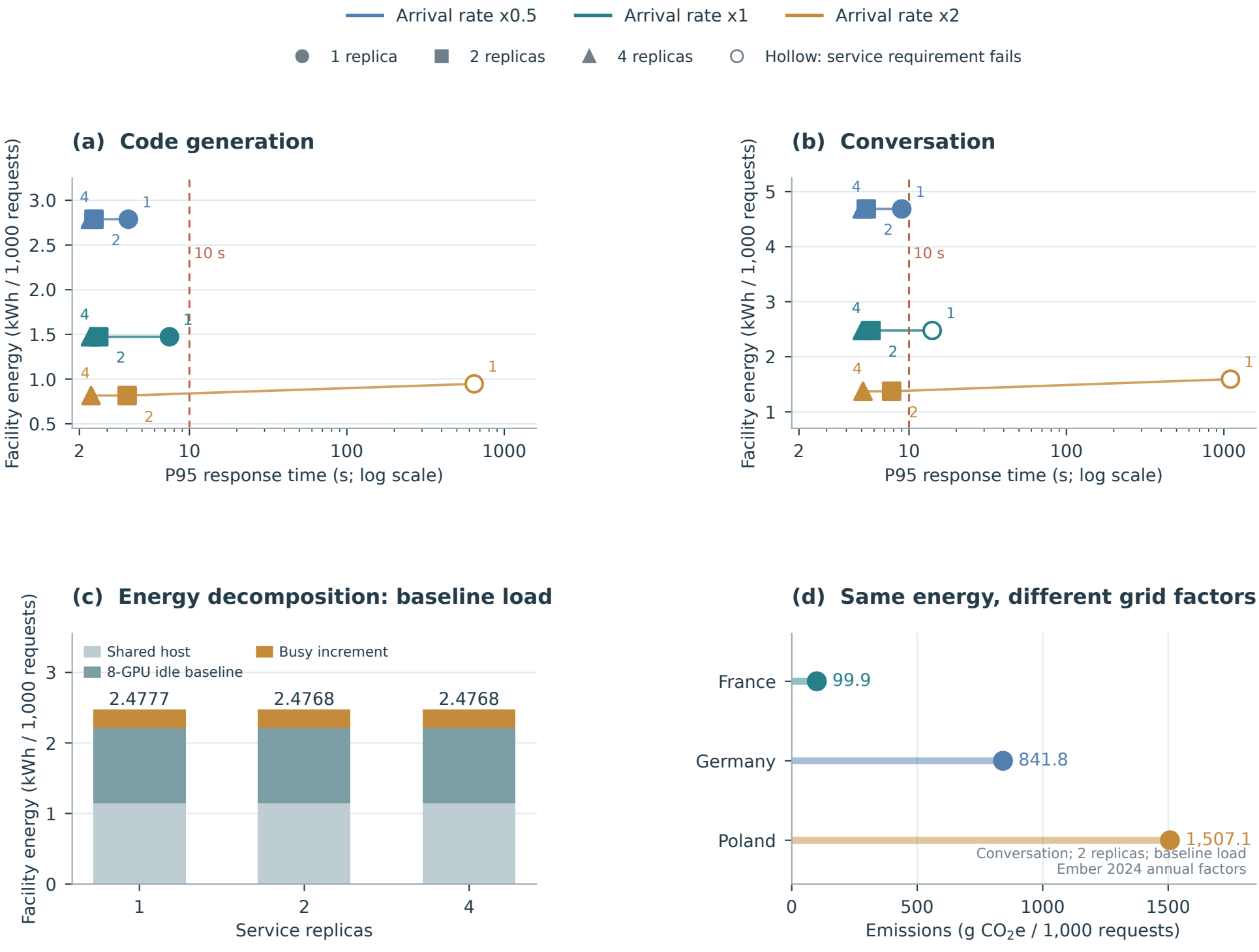
Arrival rate x0.5
Arrival rate x1
Arrival rate x2
1 replica
2 replicas
4 replicas
Hollow: service requirement fails
(a) Code generation
Facility energy (kWh / 1,000 requests)
P95 response time (s; log scale)
10 s
(b) Conversation
Facility energy (kWh / 1,000 requests)
P95 response time (s; log scale)
10 s
(c) Energy decomposition: baseline load
Facility energy (kWh / 1,000 requests)
Shared host
Busy increment
8-GPU idle baseline
2.4777
2.4768
2.4768
Service replicas
(d) Same energy, different grid factors
France
Germany
Poland
99.9
841.8
1,507.1
Conversation; 2 replicas; baseline load
Ember 2024 annual factors
Emissions (g CO2e / 1,000 requests)